

Rafi Zaman

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Electrical & Computer Engineering + Machine Learning

Expected 2028

- Relevant Coursework: LLM Systems (PhD), Parallel Computing (PhD), Logic Design & Verification, Distributed Systems, Structure & Design of Digital Systems, Data Structures & Algorithms, Object-Oriented Programming, Computer Systems, Analog Devices, Concepts of Mathematics, Mathematics of Electrical Engineering

PROFESSIONAL/ACADEMIC EXPERIENCE

Machine Learning Engineer Intern

Austin, TX

IBM

May 2026 – Present

- Trained classifiers on watsonx Orchestrate /orchestrate/runs execution logs to detect agent failures across Voice Controller, AI Gateway, and wo-proxy paths, surfacing top failure modes within the Voice Integration Team.
- Instrumented the Orchestrator Agent, which invokes tools and integrations, with 43-panel Loki + Prometheus dashboards tracking run volume, token usage, agent/tool/LLM signals, and pod CPU/memory utilization.

Machine Learning Infrastructure Intern

New York, NY

DeepSpace

January 2026 – April 2026

- Built AgentUsageTracker to monitor per-command LLM cost attribution across the agent pipeline tracking input/output token usage, applying model pricing, and logging structured billing events via API POST
- Developed D3.js visual rendering engine with neural network-based layout algorithms processing 50+ diagram types to generate dynamic SVG visualizations (neural networks, molecular structures, code flows) with sub-200ms render time

Firmware Engineering Intern

Pittsburgh, PA

Carnegie Mellon University Biorobotics

April 2025 – October 2025

- Led ESP32 embedded systems for TeleLine ventilator monitoring with real-time 3-point anomaly detection algorithm identifying 75% flow/temperature deviations and achieving 100ms HTTP POST response times
- Designed PSoC schematic (dual UART, GPIO) and C firmware detecting 10 SLM flow/2°C temperature anomalies

PROJECTS

Axiom | Python, KiCad, Triton, Reinforcement Learning

May 2026 – Present

- Architected a deterministic verification oracle that serves as the sole P/F authority for PCB designs, catching power, datasheet, and cross-domain errors invisible to standard EDA tools.
- Built a repair agent on top (verify → diagnose → propose → apply → re-verify) that auto-fixes oracle failures and rejects any regression-causing change; integrating RL across circuit design, firmware, and custom Triton GPU kernels.
- Presented oracle's benchmark results (catch rate, false-positive rate) to investors at Pareto and Entrepreneurs First.

Flash Attention | PyTorch, Triton, CUDA, Python

April 2026

- Implemented Flash Attention v2-style forward/backward GPU kernels in Triton with autotuning, avoiding full $N \times N$ attention score materialization to reduce memory usage vs. naive attention
- Built paged KV cache, grouped-query attention (GQA) with $4 \times$ smaller KV memory vs. standard multi-head attention, and prefill/decode wrappers; benchmarked decode at $\sim 28 \times$ faster than prefill at long context

Winner, BNY Mellon Track at TartanHacks | Python, Black-Scholes, FinBERT

February 2026

- Developed a Python options trading pipeline at CMU's largest hackathon (>1050 participants) that identifies mispriced opportunities by combining Black-Scholes theoretical pricing with sentiment analysis using FinBERT and VADER on real-time news from Yahoo Finance and NewsAPI.ai

RESEARCH

AdaSpec: KV-State-Aware Speculative Decoding | vLLM, Python, PyTorch, CUDA

Sep 2026 – Present

- Building an adaptive speculative decoding controller in a vLLM fork that picks draft length K per batch to maximize committed tokens/s, using the context-length distribution and KV-cache state to predict verification cost
- Designing controlled H100 experiments (fixed batch size, K , mean context; varied short/long mix) to test whether length-distribution and KV-block features choose K better than a batch-size and mean-length baseline

SecureVoI (AI Agent Safety Research, AAI submission) | Python, LLM Evaluation

July 2026

- Built SecureVoI, a framework that stops AI agents from being tricked through clarifying questions; cut unsafe actions under attack from $\sim 58\%$ to under 7% across 7 LLMs with no cost to normal performance
- Trained a TF-IDF + logistic regression classifier to detect malicious clarification responses; hit 1.000 AUROC and 97% recall on unseen attack styles, vs. 55% AUROC and under 20% recall for a keyword baseline

TECHNICAL SKILLS

Technical Skills: Python, Verilog, C++, C, Triton, APIs, ML/CV, PyTorch, CUDA, vLLM, Git, Linux, Open Source

Activities & Interests: Soccer, Carnegie Mellon Racing, Phi Delta Theta, Football, Creative Writing, Cooking